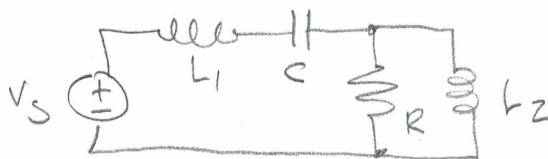


7.26)



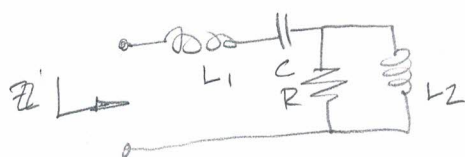
$$\begin{aligned} v_s &= 15 \cos(\omega t) \text{ V} \\ \omega &= 5 \text{ rad/sec} \\ L_1 &= 1 \text{ H}, L_2 = 2 \text{ H} \\ C &= 0.1 \text{ F} \\ R &= 5 \Omega \end{aligned}$$

Beräkna reaktiv och aktiv effekt som lasten förbrukar

Lösning: vi vet att

$$S = \frac{1}{2} V I^* = \frac{1}{2} V \frac{V^*}{Z^*} = \frac{1}{2} |V|^2 \frac{1}{Z^*} \quad (1)$$

Dvs Z^* måste beräknas



vi får

$$Z_p^{-1} = \frac{1}{R} + \frac{1}{j\omega L_2}$$

$$Z_p = \frac{j\omega L_2 R}{R + j\omega L_2} \quad (2)$$

i serie med L_1 och C

$$\begin{aligned} Z &= j\omega L_1 + \frac{1}{j\omega C} + Z_p = \\ &= j\omega L_1 + \frac{1}{j\omega C} + \frac{j\omega L_2 R}{R + j\omega L_2} \end{aligned} \quad (3)$$

med siffror

$$Z = j5 - 1 + \frac{j \cdot 5 \cdot 2 \cdot 5}{5 + j5 \cdot 2} =$$

$$= j5 - j2 + \frac{50j}{5 + j10} =$$

$$= 3j + \frac{50 \angle 90^\circ}{\sqrt{25} \angle 63.43^\circ} = 3j + 4.472 \angle 26.57^\circ \approx$$

$$\approx 3j + 3.999 + j2.000 = 3.999 + 5j =$$

$$= \sqrt{41} \angle 51.34^\circ$$

(4)

(4) & (1) numeriskt

$$S = \frac{1}{2} 15^2 \frac{1}{\sqrt{41} \angle 51.34^\circ} \approx 17.569 \angle -51.34^\circ \text{ VA}$$

DVS

$$S = P + jQ = 10.976 + j13.719 \text{ VA}$$

Aut. 1, f. 12 $\omega = 5 \text{ rad/sec}$

$$\begin{cases} P = 10.98 \text{ W} \\ Q = 13.72 \text{ VAR} \end{cases}$$

f. 12 $\omega = 15 \text{ rad/sec}$ f. 12 p. 12 s. 12

$$Z = j15 + \frac{1}{j15 \cdot 0.1} + \frac{j \cdot 15 \cdot 2 \cdot 5}{5 + j15 \cdot 2} =$$

$$= j14.33 + \frac{j150}{\sqrt{925} / 80.54^\circ} \approx j14.53 + 4.93 / 9.46^\circ \approx$$

$$\approx 4.863 + j15.340 \approx 16.092 / 72.41^\circ$$

$$\Rightarrow S = \frac{1}{2} 15^2 \frac{1}{16.092 / 72.41^\circ} = \underline{\underline{6.991 / 72.41^\circ}}$$

$$S = 2.113 + j6.664$$

DVS.

$$\begin{cases} P = 2.11 \text{ W} \\ Q = 6.66 \text{ VAR} \end{cases}$$

SVAR:

$$a) \begin{cases} P = 10.98 \text{ W} \\ Q = 13.72 \text{ VAR} \end{cases}$$

$$b) \begin{cases} P = 2.11 \text{ W} \\ Q = 6.66 \text{ VAR} \end{cases}$$